

STA 2002

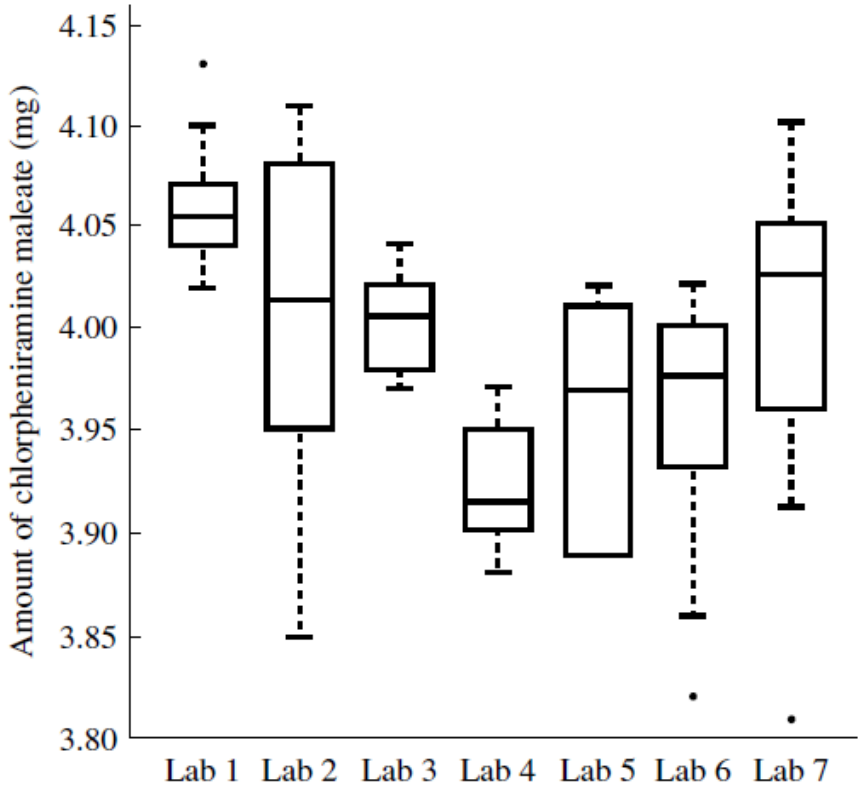
Lecture 20

One-way ANOVA

Intuition

Measurements of composites that had nominal dosages equal to 4 mg were made by seven laboratories, each laboratory making 10 measurements.

Lab 1	Lab 2	Lab 3	Lab 4	Lab 5	Lab 6	Lab 7
4.13	3.86	4.00	3.88	4.02	4.02	4.00
4.07	3.85	4.02	3.88	3.95	3.86	4.02
4.04	4.08	4.01	3.91	4.02	3.96	4.03
4.07	4.11	4.01	3.95	3.89	3.97	4.04
4.05	4.08	4.04	3.92	3.91	4.00	4.10
4.04	4.01	3.99	3.97	4.01	3.82	3.81
4.02	4.02	4.03	3.92	3.89	3.98	3.91
4.06	4.04	3.97	3.90	3.89	3.99	3.96
4.10	3.97	3.98	3.97	3.99	4.02	4.05
4.04	3.95	3.98	3.90	4.00	3.93	4.06



Are the differences in the measurements from the various labs significant, or might they be due to chance?

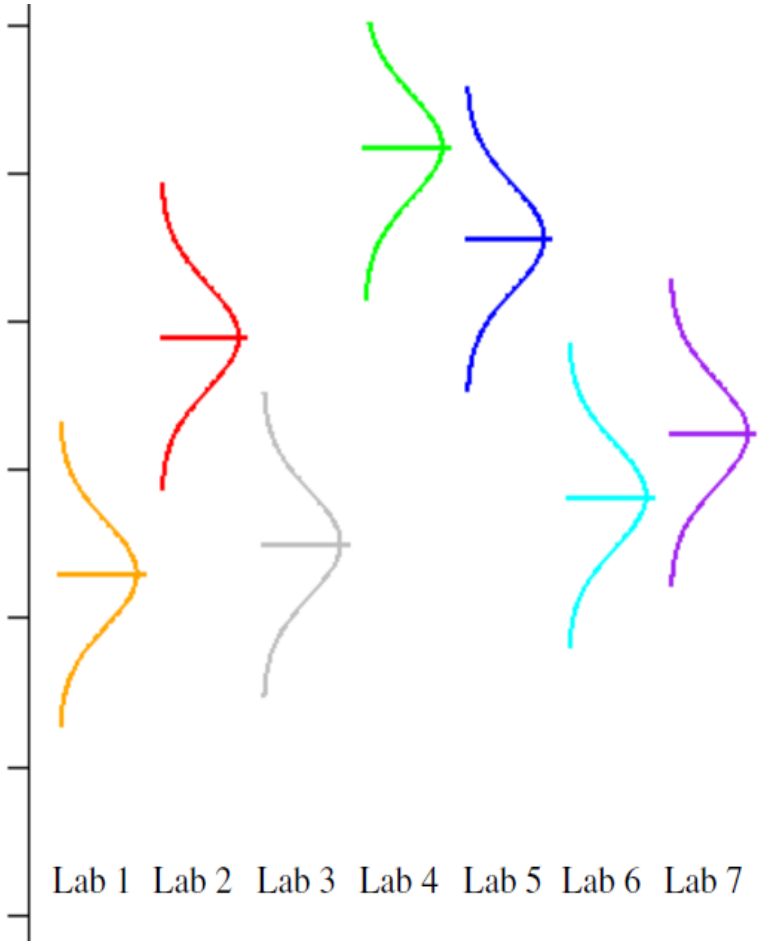
Intuition

There are two possible sources of variability in the data:

variability within labs

variability between labs





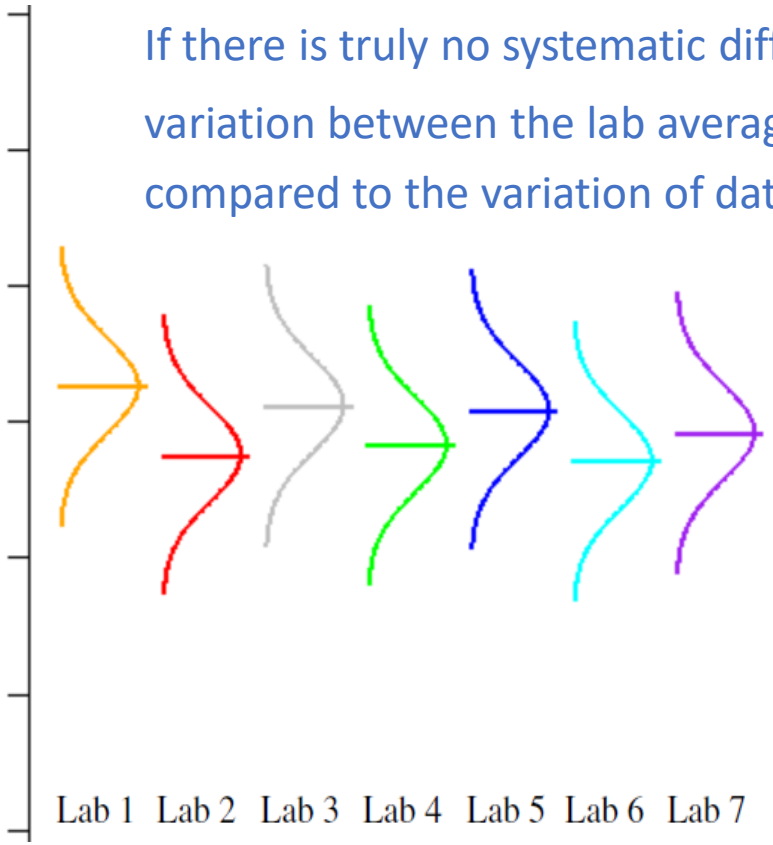
Intuition

H_0: there is no lab difference

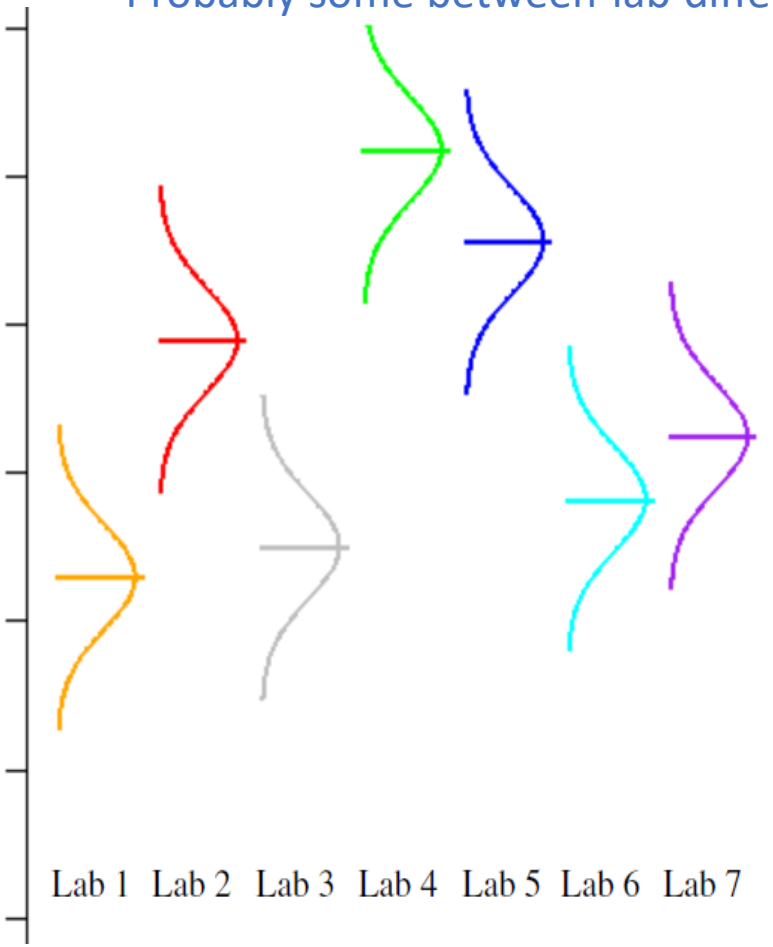
Under H_0, the variation in lab average only come from with-lab variation.

$var(X) = \sigma^2$ $\quad$ $var(\bar{X}) = \frac{\sigma^2}{n}$, n is per-group sample size

If there is truly no systematic difference between labs, the variation between the lab averages ($\frac{\sigma^2}{n}$) should be small compared to the variation of data points within labs (σ^2).



Probably some between-lab difference.



Formal setup

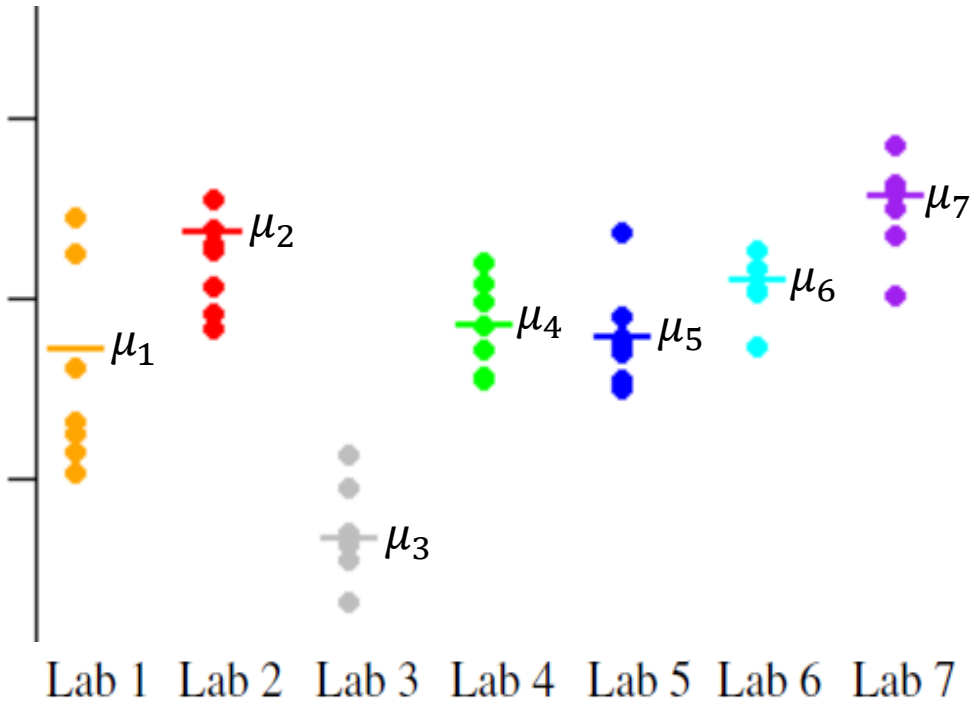
$X_{i1}, X_{i2}, \ldots, X_{in_i}$ is *i.i.d.* sample from $N(\mu_i, \sigma^2), i = 1, 2, \ldots, m$

$H_0: \mu_1 = \mu_2 = \cdots = \mu_m$

X_{ij}

index of group

index of sample within group



Mean value of the ith group

$$\overline{X}_{i\cdot} = \frac{1}{n_i} \sum_{j=1}^{n_i} X_{ij}$$

Notice we do not assume each group contains same number of samples

Formal setup

Overall mean:

$$\overline{X}_{..} = \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i} X_{ij} \qquad n = n_1 + n_2 + \cdots + n_m$$

$$\mathrm{SS(TO)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i} (X_{ij} - \overline{X}_{..})^2, \text{ the total sum of squares}$$

$$\mathrm{SS(E)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i} (X_{ij} - \overline{X}_{i.})^2, \quad \text{the sum of squares within treatments, groups, or classes, often called the error sum of squares;}$$

$$\mathrm{SS(T)} = \sum_{i=1}^{m} n_i\,(\overline{X}_{i.} - \overline{X}_{..})^2, \quad \text{the sum of squares among the different treatments, groups, or classes, often called the between-treatment sum of squares.}$$

Test statistic

$$\mathrm{SS(TO)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{..})^2$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.} + \overline{X}_{i.} - \overline{X}_{..})^2$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.})^2 + \sum_{i=1}^{m}\sum_{j=1}^{n_i}(\overline{X}_{i.} - \overline{X}_{..})^2 + 2\sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.})(\overline{X}_{i.} - \overline{X}_{..})$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.})^2 + \sum_{i=1}^{m} n_i(\overline{X}_{i.} - \overline{X}_{..})^2 + 2\sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.})(\overline{X}_{i.} - \overline{X}_{..})$$

=0

$$\mathrm{SS(TO)} = \mathrm{SS(E)} + \mathrm{SS(T)}$$

Test statistic

H_0: *All n samples are from the same distribution*

Under H_0, $\sigma_1 = \sigma_2 = \cdots = \sigma_m = \sigma$

$$\frac{\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i\cdot})^2}{\sigma^2} \sim \chi^2(n_i - 1)$$

The sum of m of these independent chi-square random variables follows a chi-squared distribution.

$$\frac{\sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i\cdot})^2}{\sigma^2} \sim \chi^2((n_1 - 1) + (n_2 - 1) + \cdots + (n_m - 1))$$

$$\frac{\mathrm{SS(E)}}{\sigma^2} \sim \chi^2(n - m)$$

Note: this doesn't require $\mu_1 = \mu_2 = \cdots = \mu_m$

Test statistic

$Under\ H_0: \mu_1 = \mu_2 = \cdots = \mu_m = \mu$

$$\frac{\sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij}-\overline{X}_{..})^2}{\sigma^2} \sim \chi^2(n-1)$$

$$\frac{\mathrm{SS(TO)}}{\sigma^2} \sim \chi^2(n-1)$$

Test statistic

Under $H_0: \mu_1 = \mu_2 = \cdots = \mu_m = \mu$

For a simple case, let us assume $n_1 = n_2 = \cdots = n_m = k$

$$\bar{X}_{i\cdot} \sim N\left(\mu, \frac{\sigma^2}{k}\right)$$

$$\frac{1}{m}\sum_{i=1}^{m} \bar{X}_{i\cdot} = \bar{X}_{\cdot\cdot}$$

This is because $k\bar{X}_{i\cdot} = \sum_{j=1}^{k} X_{ij}$

$$\frac{\sum_{i=1}^{m}(\bar{X}_{i\cdot} - \bar{X}_{\cdot\cdot})^2}{\sigma^2/k} \sim \chi^2(m-1)$$

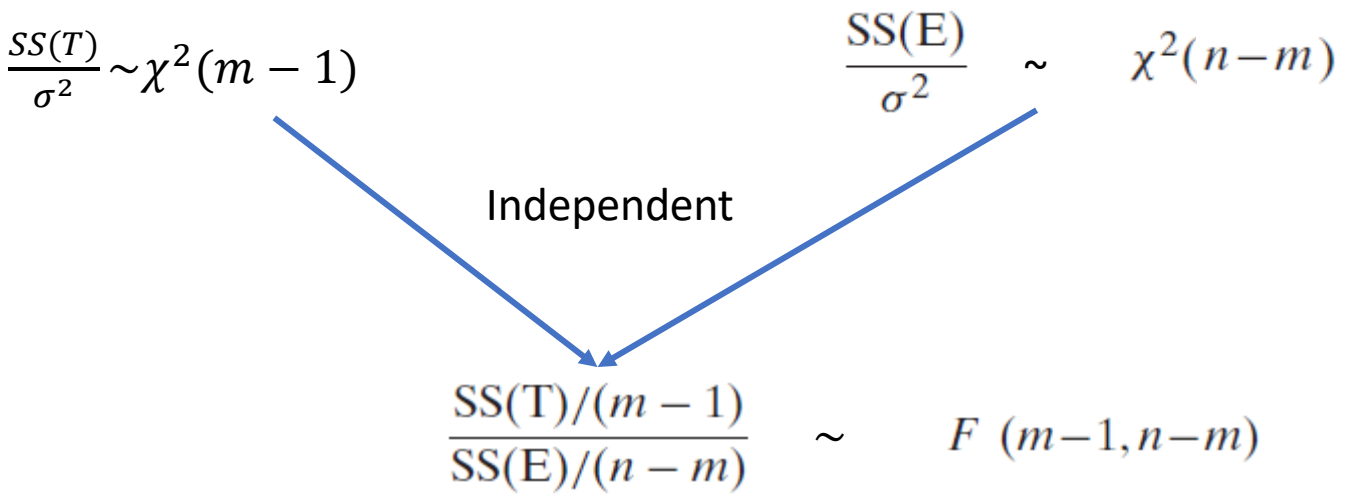
$$\frac{\sum_{i=1}^{m} k(\bar{X}_{i\cdot} - \bar{X}_{\cdot\cdot})^2}{\sigma^2} \sim \chi^2(m-1)$$

It can be shown that $\frac{\sum_{i=1}^{m} n_i(\bar{X}_{\cdot\cdot} - \bar{X}_{\cdot\cdot})^2}{\sigma^2} \sim \chi^2(m-1)$ even when we don't assume equal group size.

$$\frac{SS(T)}{\sigma^2} \sim \chi^2(m-1) \text{ under } H_0.$$

Test statistic

Under H_0



SS(T) is large when the group means are very different, relative to the within-group variation.

The critical region is of the form $F \geq F_{\alpha}(m-1, n-m)$.

Example

Table 9.3-3 Illustrative data

	Observations			$\overline{X}_{i\cdot}$
x_1:	13	8	9	
x_2:	15	11	13	
x_3:	8	12	7	
x_4:	11	15	10	
$\overline{x}_{\cdot\cdot}$				

$$\text{SS(E)} = (13-10)^2 + (8-10)^2 + \cdots + (15-12)^2 + (10-12)^2 = 50$$

$$\text{SS(T)} = 3[(10-11)^2 + (13-11)^2 + (9-11)^2 + (12-11)^2] = 30$$

$$\frac{\text{SS(T)}/(m-1)}{\text{SS(E)}/(n-m)} = \frac{30/3}{50/8} = 1.6 \quad < 4.07 \quad = F_{0.05}(3,8)$$

Example

Table 9.3-2 Analysis-of-variance table

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F Ratio
Treatment	SS(T)	$m-1$	$\text{MS(T)} = \dfrac{\text{SS(T)}}{m-1}$	$\dfrac{\text{MS(T)}}{\text{MS(E)}}$
Error	SS(E)	$n-m$	$\text{MS(E)} = \dfrac{\text{SS(E)}}{n-m}$	
Total	SS(TO)	$n-1$		

Table 9.3-3 Illustrative data

	Observations			$\overline{X}_{i\cdot}$
x_1:	13	8	9	10
x_2:	15	11	13	13
x_3:	8	12	7	9
x_4:	11	15	10	12
$\overline{x}_{\cdot\cdot}$				11

Table 9.3-4 ANOVA table for illustrative data

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F Ratio	p-value
Treatment	30	3	30/3	1.6	0.264
Error	50	8	50/8		
Total	80	11			

Test statistic

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{SS(TO)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{..})^2$$

$$SS(TO) = \sum_{i=1}^{m}\sum_{j=1}^{n_i} X_{ij}^2 - n\bar{X}_{..}^2$$

$$\text{SS(TO)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i} X_{ij}^2 - \frac{1}{n}\left[\sum_{i=1}^{m}\sum_{j=1}^{n_i} X_{ij}\right]^2$$

$$\text{SS(T)} = \sum_{i=1}^{m} n_i\,(\overline{X}_{i.} - \overline{X}_{..})^2$$

$$SS(T) = \sum_{i=1}^{m} n_i\bar{X}_{i.}^2 - n\bar{X}_{..}^2$$

This is equivalent to replacing every X_{ij} with its group mean, and then calculate the sample variance.

$$\text{SS(T)} = \sum_{i=1}^{m}\frac{1}{n_i}\left[\sum_{j=1}^{n_i} X_{ij}\right]^2 - \frac{1}{n}\left[\sum_{i=1}^{m}\sum_{j=1}^{n_i} X_{ij}\right]^2$$

$$\text{SS(E)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.})^2$$

$$SS(E) = \sum_{i=1}^{m}(n_i - 1)s_i^2$$

s_i^2 is the sample variance of the ith group

Example

Pulmonary function in the following six groups

Table 12.1 FEF data for smoking and nonsmoking males

Group number, *i*	Group name	Mean FEF (L/s)	*sd* FEF (L/s)	n_i
(1) *Nonsmokers* (*NS*)	NS	3.78	0.79	200
(2) *Passive smokers* (*PS*)	PS	3.30	0.77	200
(3) *Noninhaling smokers* (*NI*)	NI	3.32	0.86	50
(4) *Light smokers* (*LS*)	LS	3.23	0.78	200
(5) *Moderate smokers* (*MS*)	MS	2.73	0.81	200
(6) *Heavy smokers* (*HS*)	HS	2.59	0.82	200

Source: Reprinted by permission of *The New England Journal of Medicine, 302*(13), 720–723, 1980.

$$\text{SS(T)} = \left[200(3.78)^2 + 200(3.30)^2 + \cdots + 200(2.59)^2\right] - \frac{\left[200(3.78) + 200(3.30) + \cdots + 200(2.59)\right]^2}{1050}$$

$$= 10,505.58 - 3292^2/1050 = 10,505.58 - 10,321.20 = 184.38$$

$$SS(T) = \sum_{i=1}^{m} n_i \bar{X}_{i\cdot}^2 - n\bar{X}_{\cdot\cdot}^2$$

Example

Table 12.1 FEF data for smoking and nonsmoking males

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(1) *Nonsmokers* (*NS*)	NS	3.78	0.79	200
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$$SS(E) = 199(0.79)^2 + 199(0.77)^2 + 49(0.86)^2 + 199(0.78)^2 + 199(0.81)^2 + 199(0.82)^2$$
$$= 124.20 + 117.99 + 36.24 + 121.07 + 130.56 + 133.81 = 663.87$$

$$SS(E) = \sum_{i=1}^{m} (n_i - 1)s_i^2$$

$$F = \frac{184.38/5}{663.87/(1050 - 6)} = 58.0 \sim F_{5,1044} \text{ under } H_0$$

$F_{0.001}(5, 1044)=4.42 < 58.0$

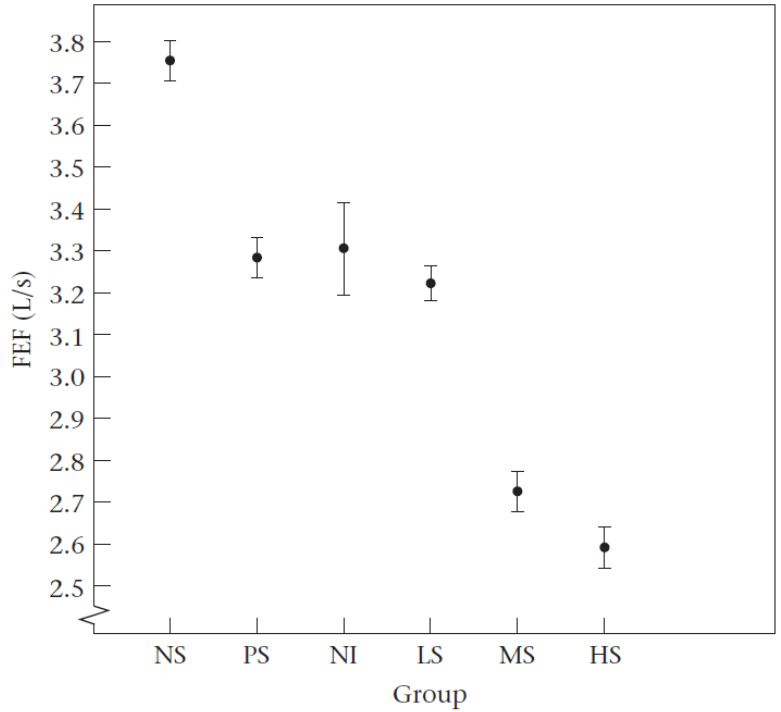
Reject H_0, that all the means are equal, and can conclude that at least two of the means are significantly different.

Example

Table 12.3 ANOVA table for FEF data in Table 12.1

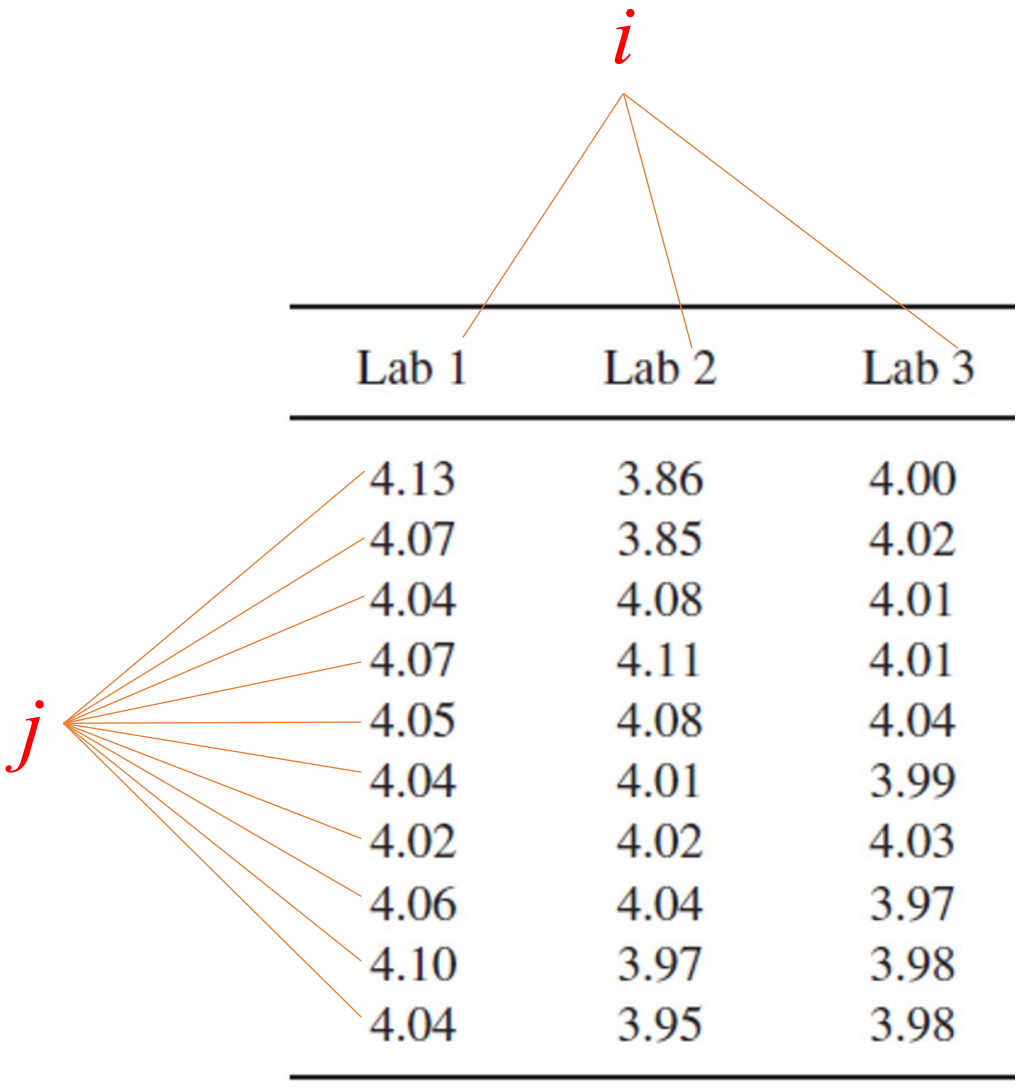
		SS	*df*	MS	*F* statistic	*p*-value
Treatment	Between	184.38	5	36.875	58.0	$p < .001$
Error	Within	663.87	1044	0.636		
Total	Total	848.25				

Figure 12.6 Mean ± *se* for FEF for each of six smoking groups



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From the data generation perspective



Lab 1	Lab 2	Lab 3
4.13	3.86	4.00
4.07	3.85	4.02
4.04	4.08	4.01
4.07	4.11	4.01
4.05	4.08	4.04
4.04	4.01	3.99
4.02	4.02	4.03
4.06	4.04	3.97
4.10	3.97	3.98
4.04	3.95	3.98

The jth observation in the ith lab

$$X_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

Overall mean
Estimated by $\bar{X}_{..}$

Effect of the ith lab
Estimated by $\bar{X}_{i.} - \bar{X}_{..}$

Error term $\sim N(0, \sigma^2)$
Estimated by $X_{ij} - \bar{X}_{i.}$

The variance decomposition

The deviation of an individual observation from the overall mean

$$X_{ij} - \bar{X}_{..} = \left(X_{ij} - \bar{X}_{i.}\right) + \left(\bar{X}_{i.} - \bar{X}_{..}\right)$$

This mapped to the variance decomposition

$$\mathrm{SS(TO)} = \mathrm{SS(E)} + \mathrm{SS(T)}$$

$$\mathrm{SS(TO)} = \sum_{i=1}^{m} \sum_{j=1}^{n_i} (X_{ij} - \overline{X}_{..})^2$$

$$\mathrm{SS(E)} = \sum_{i=1}^{m} \sum_{j=1}^{n_i} (X_{ij} - \overline{X}_{i.})^2,$$

$$\mathrm{SS(T)} = \sum_{i=1}^{m} n_i (\overline{X}_{i.} - \overline{X}_{..})^2,$$

Case Study: Donald Trump's Hair Styles (see the R example)











	A	B	C	D
1	student	gender	hairstyle	score
2	1	M	Daddy Look	60
3	2	F	Daddy Look	86
4	3	M	Daddy Look	95
5	4	F	Daddy Look	80
6	5	M	Daddy Look	75
7	6	F	Daddy Look	30
8	7	M	Daddy Look	90
9	8	F	Going Chestnut	1
10	9	M	Going Chestnut	10
11	10	F	Going Chestnut	45
12	11	M	Going Chestnut	62
13	12	F	Going Chestnut	20
14	13	M	Going Chestnut	5
15	14	F	Going Chestnut	15
16	15	M	Liza	95
17	16	F	Liza	65
18	17	M	Liza	50
19	18	F	Liza	50
20	19	M	Liza	60
21	20	F	Liza	16
22	21	M	Liza	30
23	22	F	Modern Gentleman	50
24	23	M	Modern Gentleman	80
25	24	F	Modern Gentleman	80
26	25	M	Modern Gentleman	18
27	26	F	Modern Gentleman	60
28	27	M	Modern Gentleman	85
29	28	F	Modern Gentleman	60